Guanhong Chen

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EDUCATION

Sun Yat-sen University Sep 2022 – Jun 2026

B.S. in Computer Science, GPA: 3.9/4.0 (Average: 90/100)

- China National Scholarship; First-Class Scholarship (Top 10%).
- Selected coursework: Robotics, Optimal Control, Machine Learning, Data Structures, Operating Systems, Optimization Methods, Numerical Computation, Mathematical Statistics, and Signal and System Processing.

RESEARCH EXPERIENCE

Robotics and Artificial Intelligence Researcher Sep 2024 – Present

COIR Lab, Sun Yat-sen University | Advisor: Prof. Weibing Li

Undergraduate Thesis: Event-Triggered Predefined-Time Adaptive Control of a Location-Aware Rigid-Flexible Coupled Robot

- Developed a rigid-flexible surgical robot prototype combining a Franka Emika Panda manipulator, a two-DoF continuum segment, a Hall-IMU sensing module, and a laser emitter for simulated minimally invasive ablation.
- Designed an event-triggered RBF neural network for online continuum-Jacobian estimation and formulated a constrained quadratic program incorporating remote-center-of-motion and joint-limit constraints; constructed a predefined-time, noise-tolerant Dini-RNN as the real-time QP solver.
- Integrated magnetic localization, adaptive control, and safety optimization into a closed-loop system, and evaluated sensing calibration, tracking, constraint satisfaction, disturbance rejection, and computation time through simulation and hardware experiments.
- Achieved a continuum-tip localization MAE of 0.02 mm (maximum error: 0.45 mm) after calibration, while maintaining both trajectory-tracking and RCM errors below 2 mm in physical experiments; nominated for Outstanding Undergraduate Thesis at Sun Yat-sen University (Top 5%).

Comparative Study of Model-Free Kinematic Control Strategies

- Systematically evaluated RBFNN, zeroing neural dynamics (ZND), and gradient neural dynamics (GND) for online Jacobian estimation and trajectory tracking under unknown manipulator kinematics.
- Analyzed convergence behavior using finite-time Lyapunov arguments and implemented adaptive RBF weight updates and error-coupling terms to improve convergence speed under kinematic uncertainty.
- Implemented the controllers in MATLAB/CoppeliaSim and on a 7-DoF Franka Emika Panda using C++/ROS at 1 kHz; compared tracking accuracy, convergence, robustness to sensing noise, and computation time under consistent conditions.

Independent Reproduction of MAPLE (ICRA 2022)

- Reproduced the core MAPLE pipeline as an independent learning project, implementing eight parameterized behavior primitives for 6-DoF manipulation and evaluating the reproduced system against a SAC baseline on object-assembly tasks.

Independent Study in Physics-Based Simulation and Computer Graphics Mar 2025 – Sep 2025

Self-directed study using publicly available NUS HCRL materials, DreamWorks Animation resources, and graphics-course exercises

- Studied numerical integration, physical simulation, shader programming, and dynamics through independent exercises and small implementations.
- Reproduced small-scale cloth-simulation experiments to examine numerical stability, physical plausibility, and visual realism.

PUBLICATIONS & PATENT

Journal Manuscript Jun 2026

First Author | Submitted to *IEEE Transactions on Cybernetics*

- *Event-Triggered Predefined-Time Adaptive Control of a Location-Aware Rigid-Flexible Coupled Robot Under Model Uncertainties and Safety Constraints*.
- Proposed a Hall-IMU localization method and event-triggered adaptive RBF estimator for continuum-robot kinematics, coupled with a safety-constrained QP and a predefined-time, noise-tolerant Dini-RNN solver.
- Validated the framework on a Franka–continuum prototype for minimally invasive laser ablation, demonstrating sub-millimeter calibrated localization, tracking and RCM errors below 2 mm, disturbance rejection, and real-time execution.

Conference Paper | Oral Presentation Jul 2025

G. Chen, W. Li, and Y. Li | IWACIII 2025

- *Comparative Study of Model-Free Kinematic Control Strategies for Robotic Manipulators*.
- Compared RBFNN, ZND, and GND under matched 7-DoF simulation and physical experiments, characterizing trade-offs among tracking accuracy, adaptation speed, sensing-noise robustness, and computation time.
- Deployed all three controllers on a Franka Emika Panda using a 1-kHz C++/ROS control loop and presented the study orally at IWACIII 2025.

Chinese Invention Patent Application | Application Accepted Jul 8, 2025

Applicant: Sun Yat-sen University | CNIPA Application No. 202510933703.8

- **Title:** *A Continuum Robot and Its Shape-Sensing Method*
- Inventor (3rd of 5); application formally accepted by the China National Intellectual Property Administration on Jul 8, 2025.
- Proposed a continuum robot integrating Hall-effect and IMU sensing for non-visual 3D shape perception; estimated magnetic-dipole positions through constrained optimization and reconstructed the full configuration in real time.

PROJECT LEADERSHIP & TEACHING

University-Level Undergraduate Innovation Project | Team Leader Dec 2024 – Dec 2025

Research on Key Technologies and Applications of Surgical Robots, Sun Yat-sen University

- Coordinated a four-member student team working on model-free robot control, continuum-robot shape sensing, system integration, and visual-servo module development.
- Compared ZND, RBFNN, and GND controllers on a Franka Emika Panda through simulations and preliminary experiments, focusing on tracking accuracy, convergence speed, and robustness.
- Explored Hall-effect and IMU-based 3D shape sensing for a continuum robot and participated in integrating the sensing and control modules into an experimental platform.

Teaching Assistant, C++ Programming

Sun Yat-sen University | Instructor: Prof. Hai Wan

- Supported programming instruction and problem solving after earning 99/100 in the course and demonstrating strong performance in collegiate programming competitions.

SELECTED HONORS & COMPETITIONS

- **ICPC Asia Regional Contest –** Silver Medal (68/300+), Team Leader Nov 2023
- **CCPC National Contest –** Silver Medal, Team Leader Nov 2023
- **Guangdong Collegiate Programming Contest –** Gold Medal, Team Leader May 2023
- Applied shortest-path algorithms, segment and Fenwick trees, disjoint sets, dynamic programming, and computational geometry while coordinating team strategy and debugging under contest time limits.

SKILLS

Programming:	C++, Python, MATLAB; ROS, PyTorch, OpenGL
Robotics & Control:	Franka Emika Panda, online Jacobian estimation, adaptive control, RCM-constrained control
Simulation & Design:	CoppeliaSim, SolidWorks, Blender, Unreal Engine
Methods:	RBF neural networks, ZND/GND, quadratic programming, recurrent neural dynamics, Hall-IMU sensor fusion